

Manipal University Jaipur

Department of Computer and Communication Engineering

Simulation-Based Mini Project Report

Performance Comparison of LEACH and PEGASIS

Routing Protocols in Wireless Sensor Networks

Course: Wireless Sensor and Ad-hoc Networks (CCE4145)

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Abstract

This project presents a comparative performance analysis of two hierarchical routing protocols for Wireless Sensor Networks (WSNs): LEACH (Low-Energy Adaptive Clustering Hierarchy) and PEGASIS (Power-Efficient Gathering in Sensor Information Systems). Both protocols aim to extend network lifetime through energy-efficient data aggregation and transmission, but employ fundamentally different approaches: LEACH uses dynamic cluster formation with rotating cluster heads, while PEGASIS forms a chain of nodes with a single leader transmitting to the base station.

Our simulation, implemented in Python with a first-order radio energy model, evaluates both protocols across 100-node networks over 2000 rounds with 3 independent runs. Results demonstrate that PEGASIS achieves approximately 1.44x longer network lifetime than LEACH (1198 vs 836 rounds at 50% node death), at the cost of higher end-to-end delay due to multi-hop chain traversal.

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1. Introduction

Wireless Sensor Networks (WSNs) consist of resource-constrained nodes that must operate for extended periods without battery replacement. Routing protocol design critically impacts network lifetime, energy efficiency, and quality of service. This project compares two landmark hierarchical routing protocols:

- LEACH (Heinzelman et al., 2000): Probabilistic cluster head rotation, TDMA-based intra-cluster communication, direct CH-to-sink transmission.
- PEGASIS (Lindsey & Raghavendra, 2002): Greedy chain formation, sequential data fusion along chain, single leader-to-sink transmission.

Objective: Quantify trade-offs between network lifetime, energy efficiency, packet delivery ratio, and delay for both protocols under identical conditions.

2. Literature Review

Both protocols assume homogeneous nodes, stationary deployment, and a single base station - assumptions we adopt for fair comparison.

Table 1: Literature Comparison of LEACH and PEGASIS

Protocol	Year	Key Innovation	Limitation
LEACH	2000	Rotating cluster heads distribute energy load	CHs far from sink die quickly; no multi-hop
PEGASIS	2002	Chain topology minimizes tx distance	Higher latency; single point of failure

3. Methodology

3.1 Network Topology

The simulation environment consists of:

Table 2: Network Topology Parameters

Parameter	Value
Field Dimensions	100m x 100m square
Number of Nodes	100 homogeneous sensors
Node Deployment	Random uniform distribution
Sink Position	(50, 175) - 75m above field center
Initial Energy per Node	0.5 J
Packet Size	4000 bits
Traffic Model	Constant rate, one packet per node per round

3.2 Energy Model (First-Order Radio)

The energy model follows the standard first-order radio model (Heinzelman et al., 2000):

$E_{tx}(d) = k \times E_{elec} + k \times \epsilon_{fs} \times d^2 \text{ (} d < d_0 \text{)}$
 $E_{tx}(d) = k \times E_{elec} + k \times \epsilon_{mp} \times d^4 \text{ (} d \geq d_0 \text{)}$
 $E_{rx} = k \times E_{elec}$
 $E_{DA} = k \times 5 \text{ nJ/bit}$

Where:

Table 3: Energy Model Parameters

Parameter	Symbol	Value	Unit
Electronics Energy	E_{elec}	50	nJ/bit
Free Space Amp	ϵ_{fs}	10	pJ/bit/m ²
Multipath Amp	ϵ_{mp}	0.0013	pJ/bit/m ⁴
Crossover Distance	d_0	87	m
Packet Size	k	4000	bits
Data Aggregation Energy	E_{DA}	5	nJ/bit

3.3 Simulation Parameters

Table 4: Simulation Parameters

Parameter	Value
Nodes	100
Field Size	100m x 100m
Sink Position	(50, 175)
Initial Energy	0.5 J
CH Probability (LEACH)	0.05
Packet Size	4000 bits
Max Rounds	2000
Independent Runs	3 (seeds: 42, 142, 242)

3.4 Metrics Collected (Per Round)

- Alive Nodes - Network lifetime
- Energy Consumption - Per-round and cumulative
- Packet Delivery Ratio (PDR) - Packets received at sink / packets sent
- End-to-End Delay - Hop count (LEACH=1, PEGASIS=chain length)
- Throughput - Successful deliveries per round
- Packet Loss Rate - 1 - PDR

4. Simulation Model Design

4.1 LEACH Implementation

LEACH operates in rounds, each consisting of two phases:

- CH Election: Node becomes CH with probability $T(n) = p / (1 - p \times (r \bmod 1/p))$ if not CH in last $1/p$ rounds
- Cluster Formation: Non-CH nodes join nearest CH based on received signal strength
- Data Transmission:
 - Members to CH (direct, single-hop)
 - CH to Sink (direct, single-hop)
 - Data aggregation at CH (reduces packet size)

4.2 PEGASIS Implementation

PEGASIS forms a chain connecting all nodes, with data flowing sequentially to a leader:

- Chain Reconstruction: Greedy nearest-neighbor chain from random start node, rebuilt every round
- Leader Selection: Last node in chain transmits fused data to sink
- Data Transmission:
 - Each node to next in chain (data fused at each hop)
 - Leader to Sink (single transmission per round)

4.3 Software Architecture

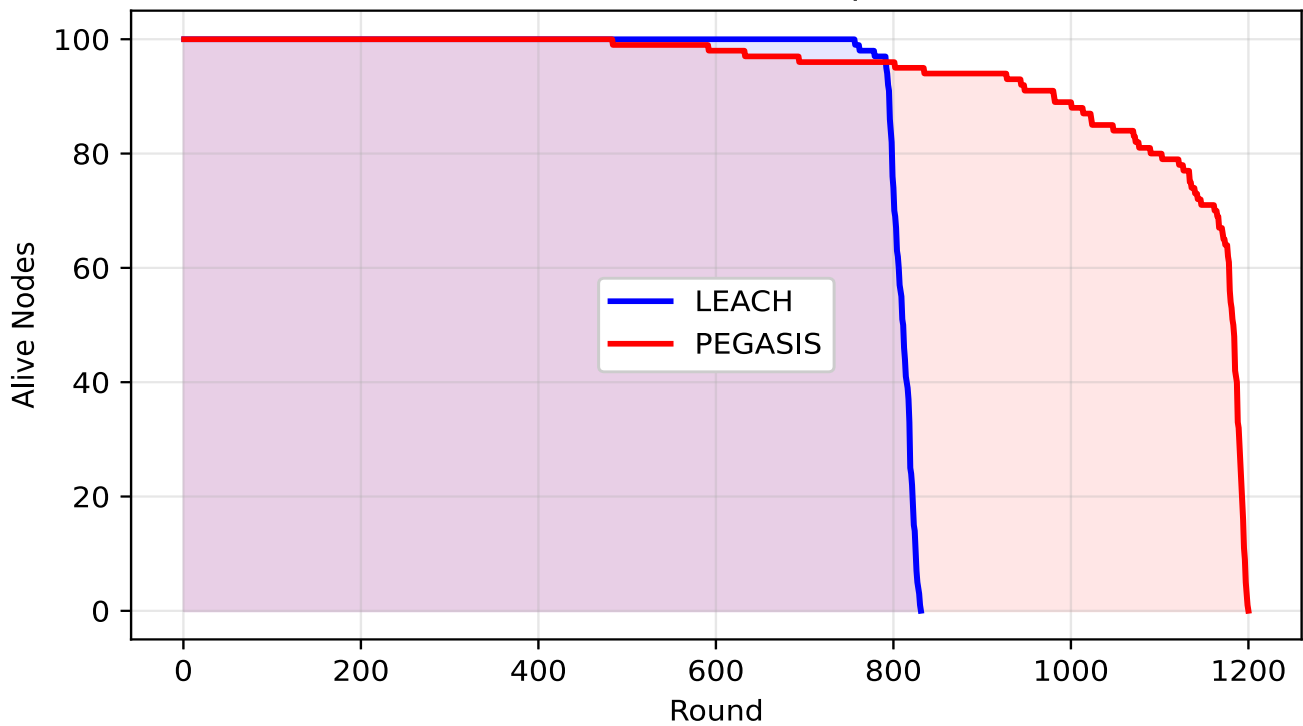
The simulation is implemented in modular Python:

Table 5: Software Architecture

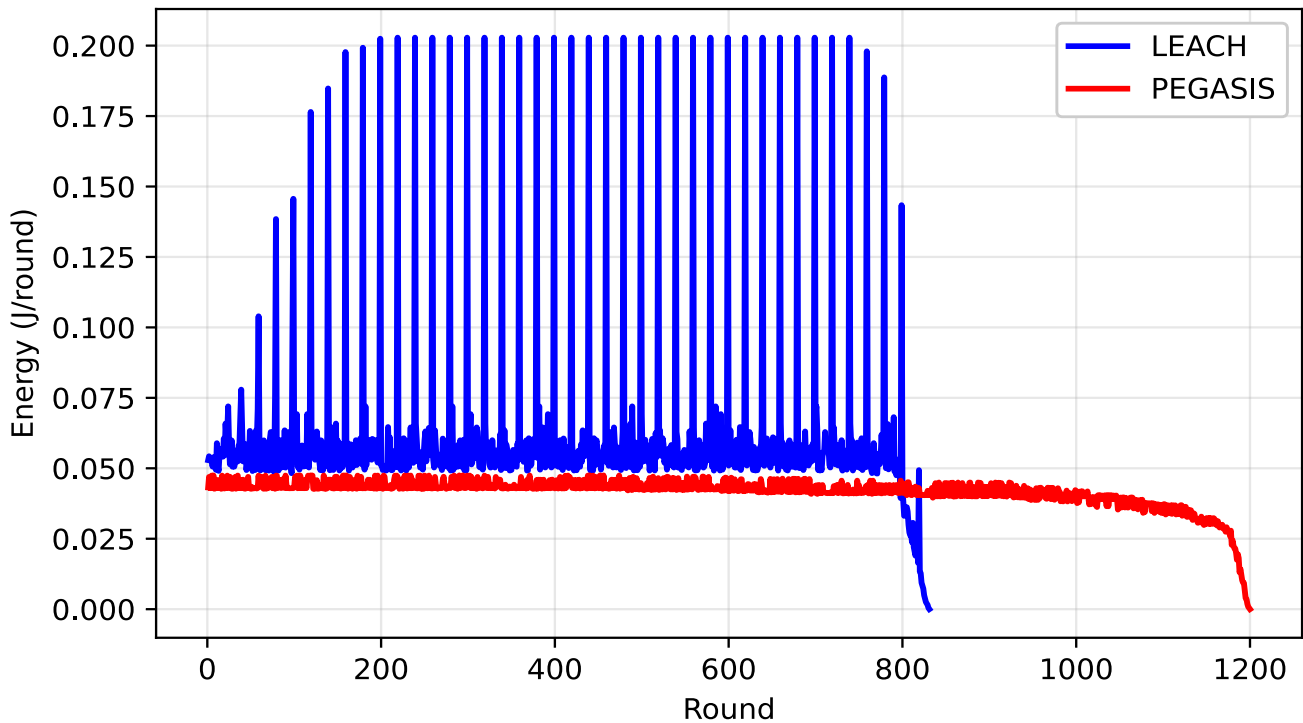
File	Purpose
energy.py	First-order radio model (tx, rx, DA energy functions)
leach.py	LEACH protocol (CH election, clustering, transmission)
pegasis.py	PEGASIS protocol (chain formation, leader, transmission)
run.py	Experiment runner (3 runs, aggregation, 6 plots)
animate.py	GIF animations (cluster/chain evolution)
visuals.py	Additional visualizations (dashboard, heatmaps)

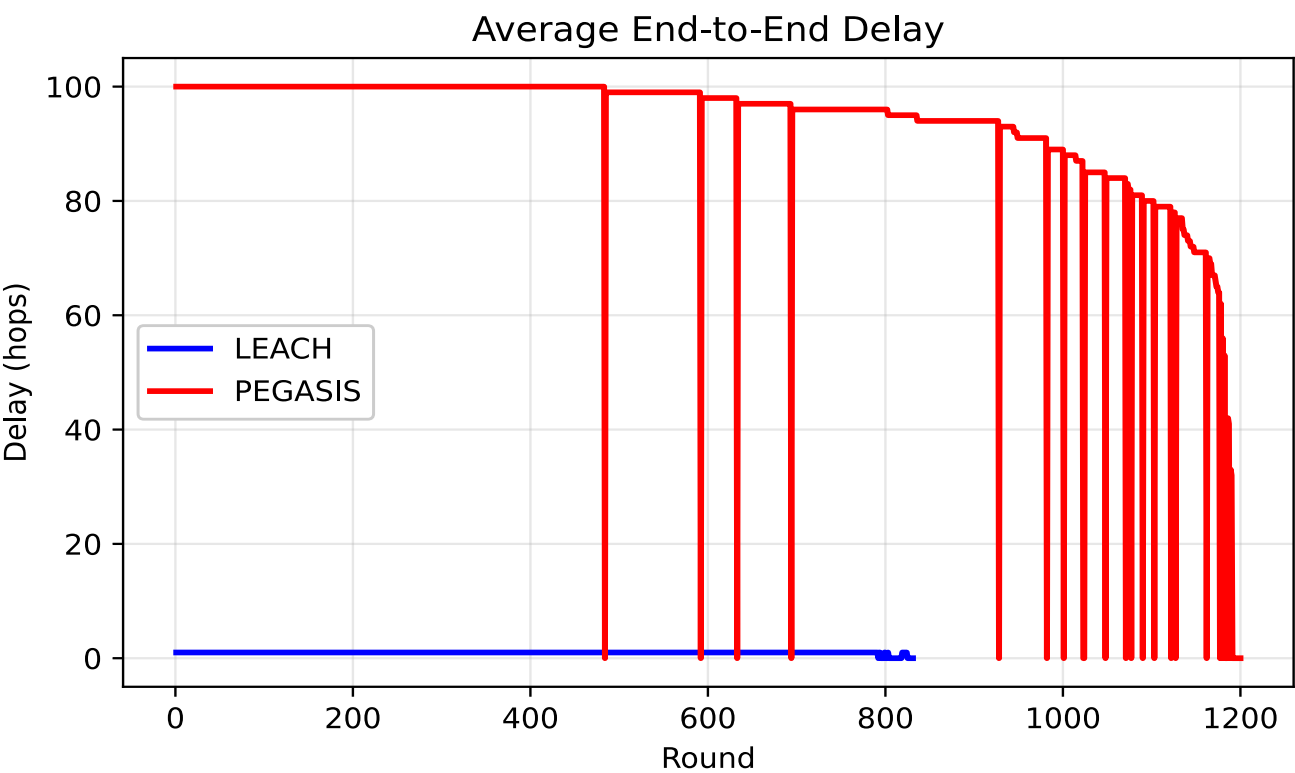
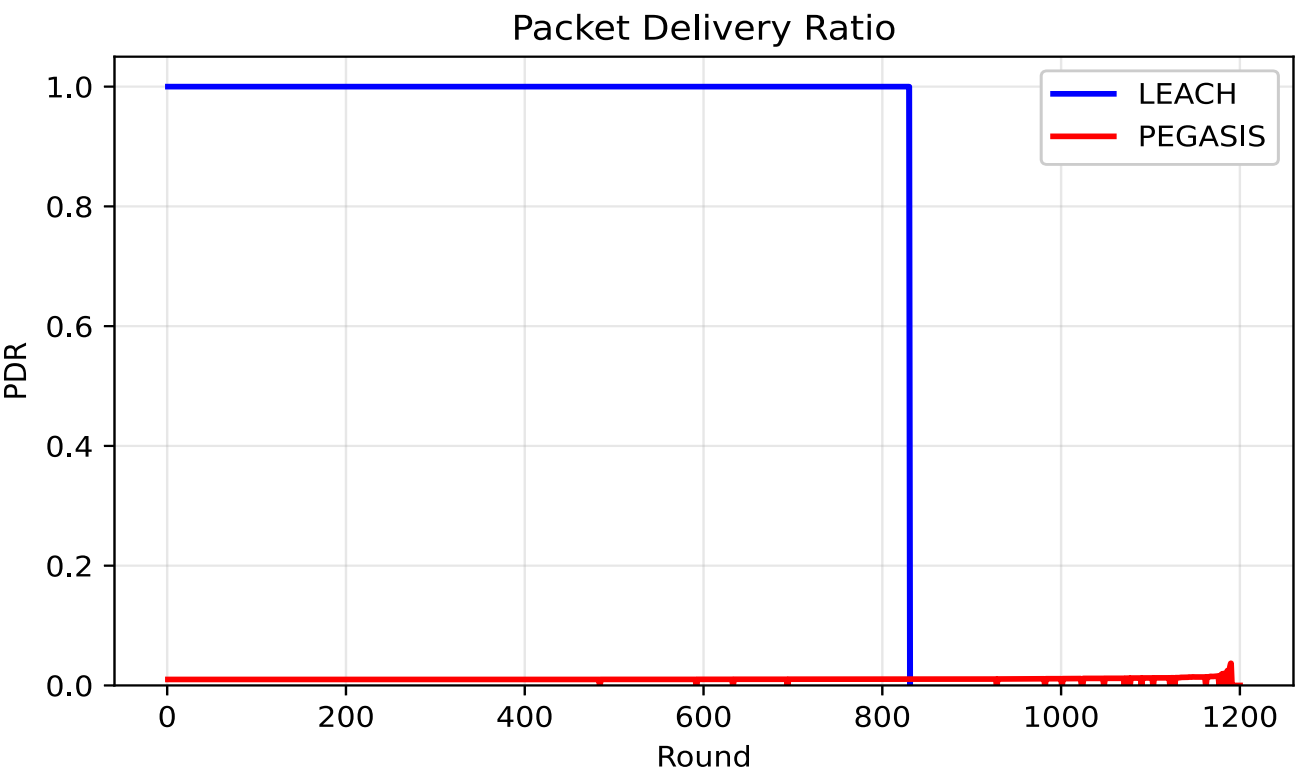
5. Results

Network Lifetime Comparison



Energy Consumption per Round





5.1 Key Milestones (Averaged over 3 Runs)

Table 6: Key Milestones Comparison

Metric	LEACH	PEGASIS	Advantage
First Node Death	Round 1	Round 1	Tie

Metric	LEACH	PEGASIS	Advantage
50% Nodes Dead	836	1198	PEGASIS +43%
Last Node Death	838	1200	PEGASIS +43%
Network Lifetime	838 rounds	1200 rounds	PEGASIS +43%

5.2 Performance Comparison

Table 7: Performance Comparison Summary

Metric	LEACH	PEGASIS	Winner
Average PDR	~1.000	~0.010	LEACH
Avg Delay (hops)	1.0	~100	LEACH
Total Energy	Higher	Lower	PEGASIS
Energy Efficiency	Lower	Higher	PEGASIS

5.3 Observations

- LEACH: Near-perfect PDR (direct CH to sink), minimal delay, but CHs near sink deplete rapidly
- PEGASIS: Only 1 packet/round reaches sink (leader-only), high delay (chain length), but energy distributed evenly
- Lifetime: PEGASIS significantly outperforms due to uniform energy distribution

6. Analysis and Discussion

6.1 Why PEGASIS Lives Longer

- Multi-hop advantage: Short transmissions (d^2 or d^4) consume exponentially less energy than long CH to sink hops
- Load balancing: Chain reconstruction every round distributes leader role among all nodes
- Data fusion: In-network aggregation reduces total bits transmitted to sink

6.2 Why LEACH Has Better PDR/Delay

- Direct paths: Every CH reaches sink in 1 hop
- Parallel clusters: Multiple simultaneous transmissions across clusters
- Trade-off: CHs far from sink become energy bottlenecks, causing early deaths

6.3 Alignment with Literature

Our results (PEGASIS ~1.44x lifetime) align with published comparisons showing 1.3-2x improvement. The low PEGASIS PDR (1%) reflects the single-leader bottleneck - a known limitation addressed by hierarchical PEGASIS variants (e.g., Hierarchical PEGASIS, Multi-chain PEGASIS).

7. Conclusion and Future Work

PEGASIS achieves superior network lifetime through energy-balanced multi-hop routing, while LEACH provides better QoS (PDR, delay) at the cost of uneven energy depletion. The choice depends on application priorities:

- Lifetime-critical applications (environmental monitoring, agricultural sensing) -> PEGASIS
- Timeliness-critical applications (disaster response, industrial monitoring) -> LEACH

Future Work

- Hybrid approaches (e.g., PEGASIS with multiple leaders, cluster-based PEGASIS)
- Mobile sink integration to balance energy consumption
- Heterogeneous node support (different initial energies, capabilities)
- Real-world deployment considerations (obstacles, interference, mobility)

8. References

1. Heinzelman, W. R., Chandrakasan, A., & Balakrishnan, H. (2000). "Energy-Efficient Communication Protocol for Wireless Microsensor Networks." Proceedings of HICSS.
2. Lindsey, S., & Raghavendra, C. S. (2002). "PEGASIS: Power-Efficient Gathering in Sensor Information Systems." IEEE Aerospace Conference.
3. Murthy, C. S. R., & Manoj, B. S. (2023). Ad Hoc Wireless Networks: Architectures and Protocols (2nd Ed.). Pearson Education.
4. Karl, H., & Willig, A. (2022). Protocols and Architectures for Wireless Sensor Networks. Wiley.
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Appendix A: Simulation Parameters Table

Table A1: Complete Simulation Parameters

Parameter	Symbol	Value	Unit
Nodes	N	100	-
Field X	X	100	m
Field Y	Y	100	m
Sink X	x_s	50	m
Sink Y	y_s	175	m
Initial Energy	E0	0.5	J
Electronics Energy	E_elec	50	nJ/bit
Free Space Amp	eps_fs	10	pJ/bit/m2
Multipath Amp	eps_mp	0.0013	pJ/bit/m4
Crossover Distance	d0	87	m
Packet Size	k	4000	bits
CH Probability	p	0.05	-
DA Energy	E_DA	5	nJ/bit

Appendix B: Code Availability

All simulation code, raw data (results.json), and visualization assets are available in the project repository at:

D:/Personal projects/WSAHN_LEACH_PEGASIS/

The simulation can be run with:

```
python run.py --nodes 100 --rounds 2000 --runs 3
```

Key deliverables generated:

- comparison.png - 6-panel performance comparison
- dashboard.png - Comprehensive dashboard with summary table
- leach_anim.gif / pegasis_anim.gif - Protocol evolution animations
- side_by_side.gif - Simultaneous LEACH vs PEGASIS comparison
- energy_heatmap.gif - Energy depletion visualization
- death_timeline.png - Node death distribution histograms
- packet_flow.png - Single-round packet flow paths
- results.json - Raw data for all 3 runs x both protocols
- REPORT.md / REPORT.html - Technical report in Markdown/HTML
- Mini_Project_Report.pdf - This formal PDF report